

GCI WORLD 2026 APRIL • FINAL ASSIGNMENT • PoC PROPOSAL

Turning Customer Data into Retained Revenue

A machine-learning churn-prevention proposal for Company A (Telecom)

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Executive Summary

Company A loses nearly half its customers. A data-driven, ML-targeted retention program can recover ~\$6M in revenue at almost no incremental cost.

49.6%

Customer churn rate

0.69

Model ROC-AUC

1.67x

Lift in top 5% at-risk

\$6.0M

Net benefit (24-mo)

- **Problem:** ~49.6% of ~100,000 customers churn within the 31–60 day window — a major recurring revenue leak.
- **Solution:** A monthly churn-risk scoring engine ranks customers so the retention team targets the highest-risk segment first.
- **Evidence:** Gradient Boosting (ROC-AUC 0.69) identifies churners 1.5–1.7× better than random targeting.
- **Impact:** Targeting the top 20% yields ~\$6.0M net benefit and ~\$2.0M more than untargeted outreach.

Market Context: Why Churn Is the Battleground

- **US telecom is a ~\$650B market** [1] — mature, saturated, and fiercely competitive.
- **Annual churn averages 20–30%** across wireless carriers [2]; acquisition is largely zero-sum.
- **Acquiring a new customer costs 5–7×** more than retaining an existing one [3].
- **A 5% lift in retention can raise profits 25–95%** (Reichheld & Sasser) [4].
- **Implication:** with little room to grow the pie, defending the existing base is the highest-ROI lever available to Company A.

THE RETENTION ECONOMICS

5–7×

cost to acquire vs. retain a customer

25–95%

potential profit increase from a
5% improvement in retention

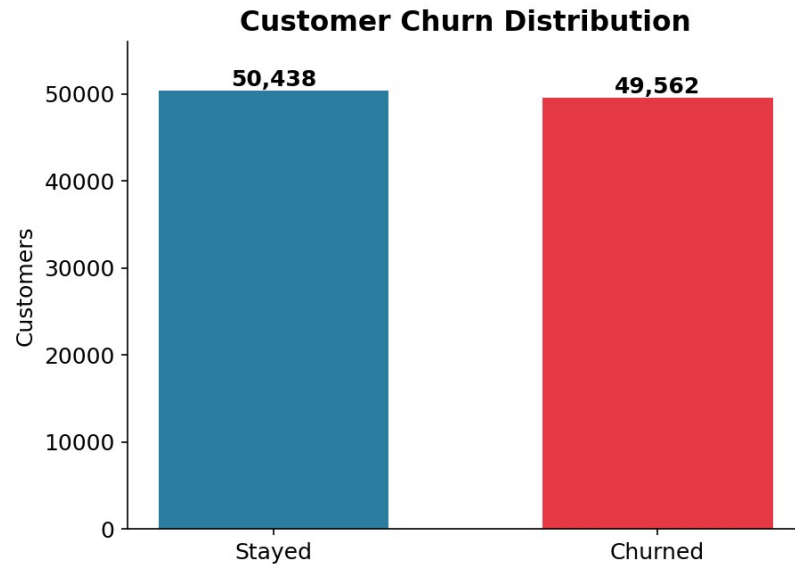
The Data We Were Given

- **Two tables, joined on Customer_ID:**
 - • **Client.csv** — 100,000 customers × 50 demographic/account columns.
 - • **Record.csv** — 100,000 rows × 51 usage/behavioral columns.
- **Merged view:** 100,000 customers × 100 features.
- **Target variable:** churn (1 = left within 31–60 days, 0 = stayed).
- **Real-world messiness:** many columns have missing values and ambiguous definitions — handled explicitly in preprocessing.

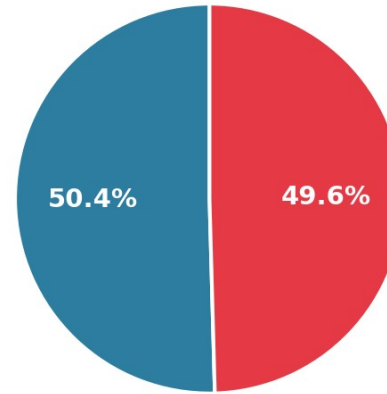
FEATURE FAMILIES

- **Usage:** minutes of use, calls placed/received, peak vs off-peak
- **Revenue:** monthly charge, overage, billing-adjusted totals
- **Service quality:** dropped / blocked / unanswered calls
- **Equipment:** handset price, equipment age, web capability
- **Account:** tenure (months), # subscribers, credit class
- **Demographics:** income, region, dwelling, household

Finding 1: Half the Base Is Walking Out



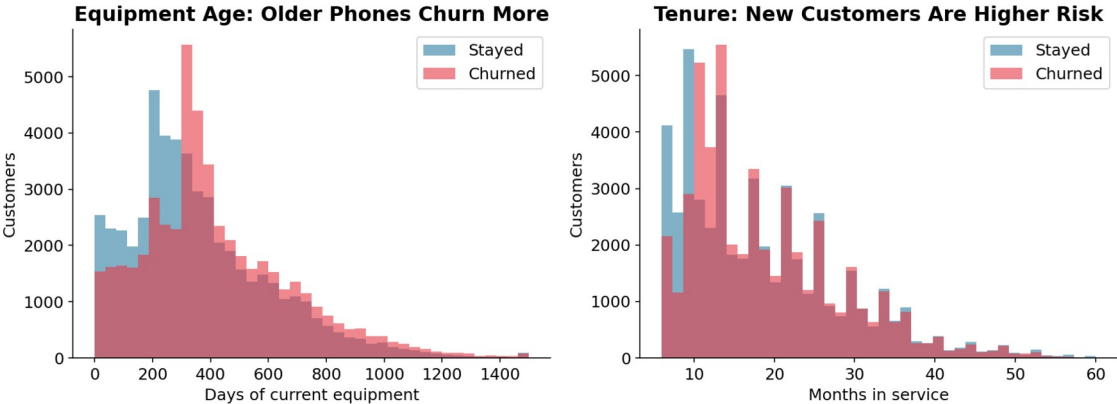
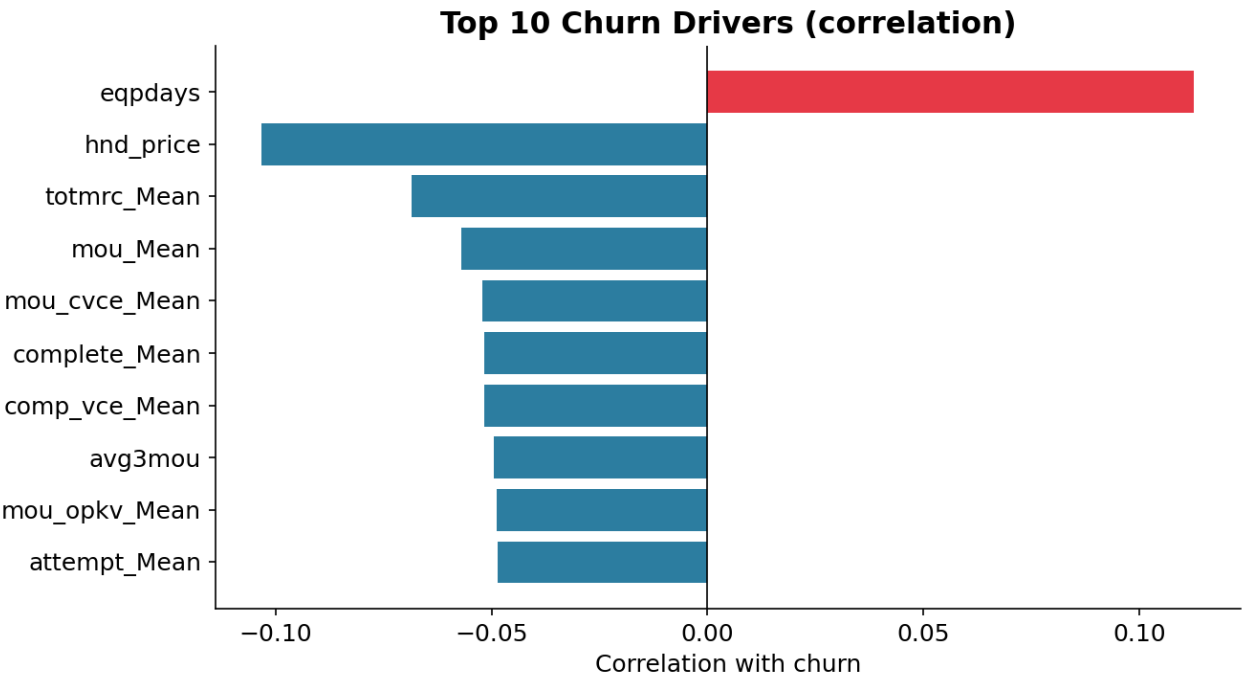
Nearly Half of Customers Churn (49.6%)



SO WHAT?

- 49.6% churn is extreme — almost a coin-flip on every customer.
- Even a small, well-targeted reduction protects millions in recurring revenue.
- The base is balanced, so the modeling challenge is ranking risk, not finding a rare event.

Finding 2: What Separates Churners From Stayers



Equipment age and handset price are the strongest signals: customers on older, cheaper phones — and those with declining usage — are the most likely to leave. These map directly to actionable retention offers (device upgrades, loyalty perks).

Problem Definition & ML Task

BUSINESS OBJECTIVE

Reduce revenue lost to churn by identifying — in advance — which customers are about to leave, so retention can act before they do.

MACHINE-LEARNING TASK

Binary classification of churn, optimized to RANK customers by risk probability rather than to label them yes/no.

Why this framing?

- **Target = churn** — directly ties to the client's core profitability problem (validated in EDA).
- **Metric = ROC-AUC** — measures how well we RANK risk; insensitive to threshold, ideal when the plan is to contact a top-N list.
- **Ranking > hard labels** — the retention budget is finite, so we prioritize the customers most likely to churn AND worth saving.

Modeling Approach

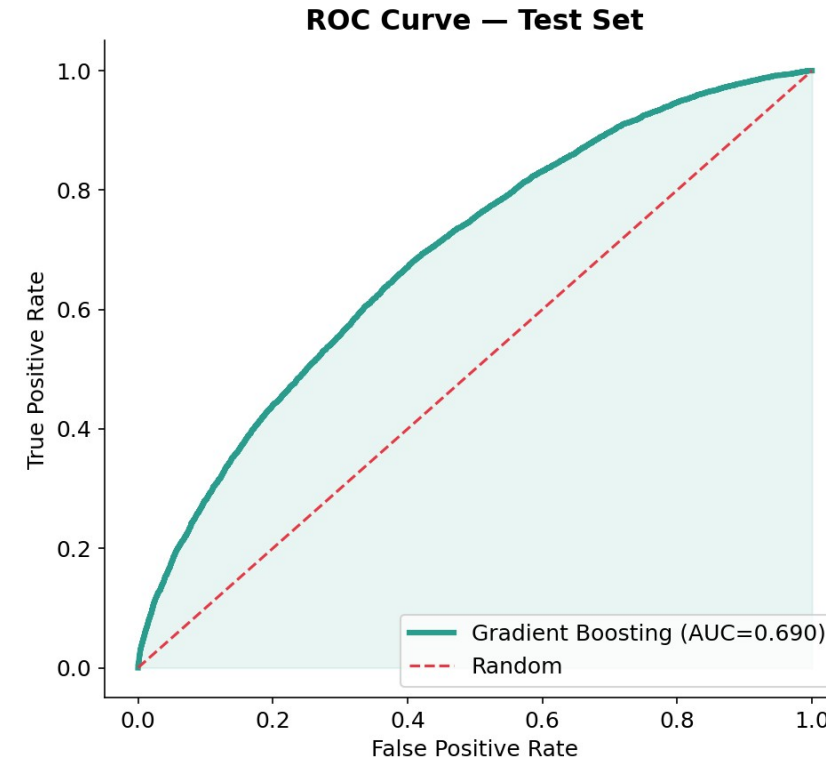
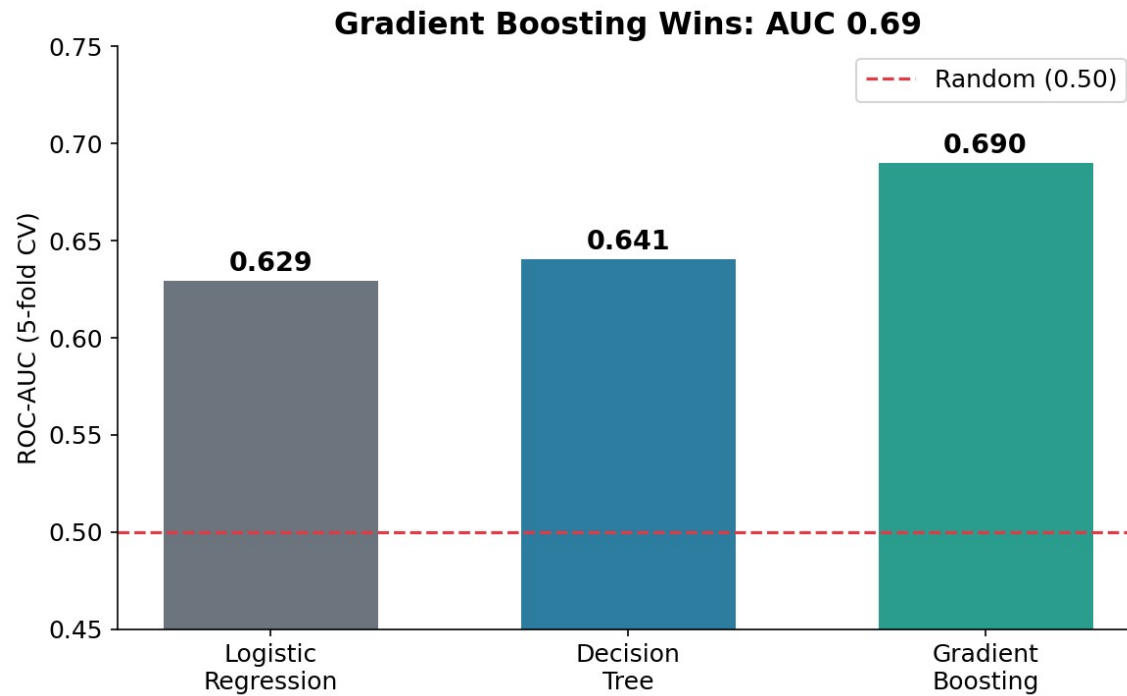
- **Preprocessing pipeline (leak-free):**
 - • **Numeric** — median imputation + standardization.
 - • **Categorical** — most-frequent imputation + one-hot encoding.
 - • **Dropped** high-cardinality / identifier columns.
- **Feature engineering:**
 - • **failure_ratio** — dropped+blocked ÷ attempted calls (service quality).
 - • **rev_per_minute** — revenue efficiency.
 - • **custcare_intensity** — support-call rate (dissatisfaction proxy).
- **Validation:** stratified 80/20 split + 5-fold cross-validation, fixed random seed (reproducible).

MODELS COMPARED

- **Logistic Regression** — interpretable, transparent baseline.
- **Decision Tree** — captures simple non-linear rules.
- **Gradient Boosting** — ensemble of trees; strongest on tabular telecom data.

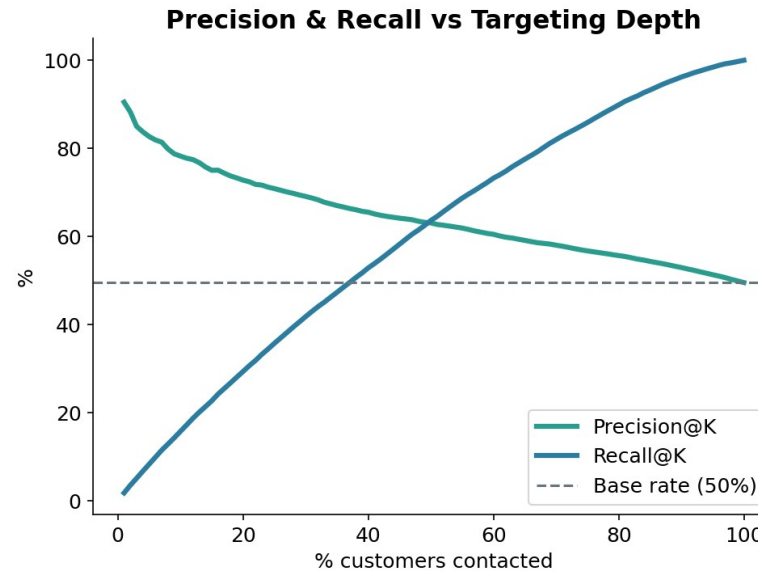
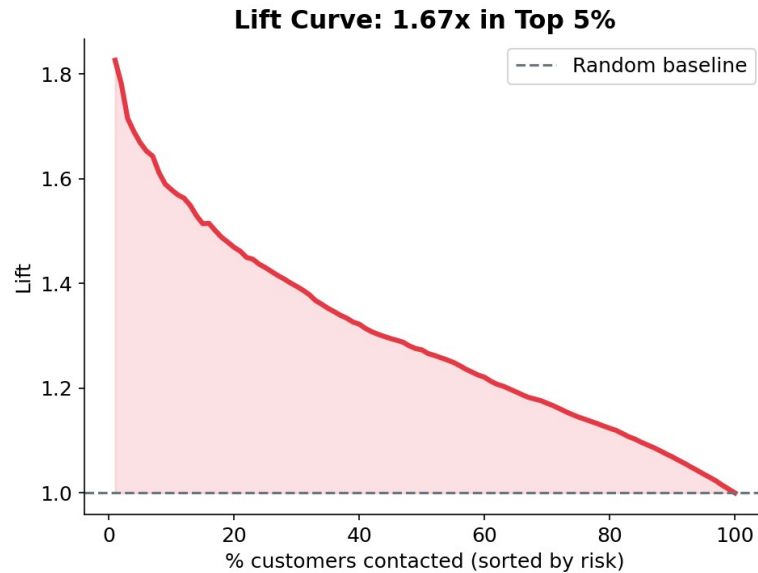
Each model handles class balance and is scored on identical folds, so the comparison is apples-to-apples.

Results: Gradient Boosting Is the Clear Winner



Model: Gradient Boosting Classifier • Metric: ROC-AUC • Score: 0.69 (test) — ~38% better than a random baseline (0.50).

From Prediction to Action: Who to Call First



READING THE CHART

- **Top 5%:** 83% are real churners — 1.67× better than random.
- **Top 20%:** 73% precision, capturing ~29% of all churners.
- **Takeaway:** a small, focused campaign captures a large share of churn risk cheaply.

The Proposal: A Monthly Churn-Risk Engine

1. SCORE

Every month, score all customers 0–1 for churn risk using the Gradient Boosting model.

2. RANK & TARGET

Send the retention team a ranked list; focus on the top 20% highest-risk customers.

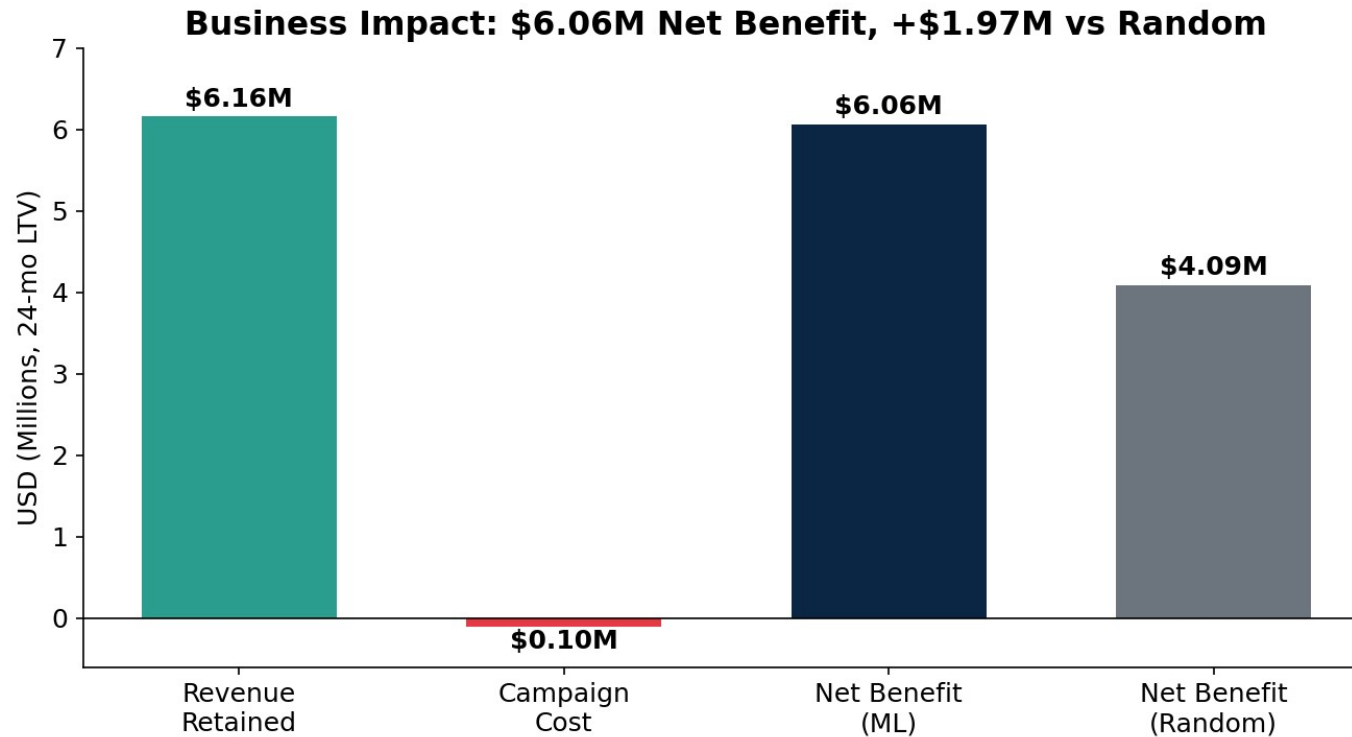
3. INTERVENE

Trigger tailored offers — device upgrades, loyalty discounts, proactive support — by risk driver.

Why it works

- **Targets the right people** — EDA-validated drivers (old handsets, declining usage) map to concrete offers.
- **Fits existing operations** — plugs into the current retention team; no new infrastructure needed for the PoC.
- **Spends efficiently** — contacting 20% instead of 100% cuts campaign cost 5× while keeping most of the upside.

Quantified Impact



TOP-20% CAMPAIGN

- 20,000 customers contacted
- 73% precision → 14,560 churners reached
- 30% save rate → ~4,368 retained
- \$58.72 avg monthly revenue / customer
- 24-month lifetime-value horizon

\$6.0M net benefit • +\$2.0M vs. random

Next Steps & Risks

Phased roadmap

- **Phase 1 (0-1 mo):** A/B test the top-20% campaign on a hold-out segment to measure true save-rate.
- **Phase 2 (1-3 mo):** Feed results back; tune offers by churn driver; add SHAP for per-customer explanations.
- **Phase 3 (3-6 mo):** Automate monthly scoring; try XGBoost/LightGBM & uplift modeling; retrain to counter drift.

Risks & mitigations

- **Model drift** → scheduled monthly retraining.
- **Save-rate uncertainty** → validate via controlled A/B test before scaling.
- **Offer cannibalization** → cap discounts; target by risk driver, not blanket.
- **Data quality / privacy** → governance review; the model uses only provided, consented data.

References

- [1] IBISWorld / Statista (2024). U.S. Wireless Telecommunications Carriers — Market Size.
- [2] McKinsey & Company. Reducing churn in telecom through advanced analytics.
- [3] Harvard Business Review. The Value of Keeping the Right Customers (Gallo, 2014).
- [4] Reichheld, F. & Sasser, W. E. Zero Defections: Quality Comes to Services, HBR; Bain & Company.
- [5] Dataset: Company A (anonymized telecom) PoC dataset — GCI World 2026 course materials.
- [6] Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12.
- [7] Friedman, J. (2001). Greedy Function Approximation: A Gradient Boosting Machine. Annals of Statistics.
- [8] Course lecture notebooks, Matsuo-Iwasawa Lab (GCI World 2026 April).

Generative AI (Claude, Anthropic) was used to assist with structuring and visualization; chat URL listed in the Omnicampus References field.